A COMPARISON OF LLM PERSONALITY PROFILES

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ABSTRACT

This study compares personality self-ratings from 55 large language model (LLM) configurations through repeated assessments with the fifty-item International Personality Item Pool questionnaire (IPIP-50). Twenty of the 22 complete profiles had higher Agreeableness, Conscientiousness, and Intellect/Imagination scores than Extraversion. Some between-model score orderings persisted across recorded response groups, while others reversed. Missing answers restricted trait coverage and left some comparisons unresolved.

Keywords: large language models, personality profiles, Big Five, IPIP-50, self-report, response coverage

1 INTRODUCTION

Language models can describe themselves as cooperative, organized, reserved, or imaginative. Here, a personality profile means the available questionnaire scores elicited by an instruction. These self-ratings do not establish psychological traits; they may reflect assistant style or learned conventions. Human experiences may receive metaphorical ratings or be marked inapplicable.

Earlier studies elicited personality-like profiles from language models [4; 5], but answers depend on wording, response options, and testing occasion [6; 7]. Broader studies found model- and prompt-dependent evidence of reliability and validity [8].

This study compares 55 configurations under one IPIP-50 assistant-self instruction. It examines recurring profiles, variation across repeats, and incomplete answers, with sensitivity checks for response selection and scoring. Because generation settings vary by model, the comparisons concern the tested configurations.

2 RELATED WORK

This selective narrative review focuses on primary studies of questionnaire responses, measurement consistency, and alternative assessments. Table 1 summarizes their main differences.

2.1 DIRECT PERSONALITY QUESTIONNAIRES

Miotto, Rossberg, and Kleinberg administered HEXACO-60 and the Human Values Scale to GPT-3 [4]. Some scores resembled human reference values, but several changed with generation temperature, and correlations among scores did not consistently match human data. Personality items were presented separately; the values study also tested prompts containing previous questions and answers.

Jiang et al. introduced the Machine Personality Inventory (MPI), with 120-item and 1,000-item versions drawn from IPIP-derived and related Big Five material [5]. They used personality prompts and vignette tests to examine assigned traits. The present study instead uses the fifty-item inventory to elicit assistant self-descriptions without assigning a target personality.

IPIP-50 provides public item wording and a published scoring key [1; 2; 3]. Its five domains are Extraversion, Agreeableness, Conscientiousness, Emotional Stability, and Intellect/Imagination. Higher Emotional Stability broadly corresponds to lower Neuroticism. Intellect/Imagination covers

vocabulary, ideas, and imagination, rather than performance on an intelligence test. These domain labels do not make scores interchangeable with other Big Five inventories.

2.2 Consistency and validity

Shu et al. converted material from 39 instruments into 693 yes/no questions and evaluated 17 models [6]. Formatting, answer labels, option order, and negation could change responses. Most models were more consistent under option-order or answer-label changes than under reversed statement meanings. A tendency to answer positively could inflate option and order consistency.

Bodroža et al. tested seven models twice, five days apart for each model, using instruments including the 60-item Big Five Inventory-2 and HEXACO-100 [7]. GPT-3 and the other models were tested in different collection periods. Models were asked to pretend to be human. Agreement varied across models, instruments, and traits. Their temporal, human-role design differs from the short-run assistant-self assessments used here.

Serapio-García et al. evaluated 18 models using the 300-item IPIP-NEO, the 44-item Big Five Inventory, and tests of reliability and validity [8]. Items were assessed separately while biographic descriptions, instructions, and response framing varied. Some larger and instruction-tuned models showed stronger evidence of reliable measurement and relationships with downstream responses. Their prompt variation, cross-instrument comparisons, and downstream tests go beyond the fixed-prompt repeats used here.

2.3 Interviews and scenario-based tests

InCharacter evaluates fidelity to assigned characters through psychological interviews [9]. Its benchmark covers 32 characters and 14 scales. LLM assessors convert interview responses into scale options or expert-style ratings, using human-perceived character labels as references.

TRAIT contains 8,000 hypothetical scenario questions derived from Big Five and Short Dark Triad material [10]. It examines sensitivity to prompts, option order, and paraphrasing. These are hypothetical choices rather than observations of deployed behavior.

Ma et al.'s Core Sentiment Inventory summarizes word associations as optimism, pessimism, and neutrality [11]. Neutrality includes inconsistent or reluctant responses as well as neutral ones. This preprint measures a different target from Big Five traits.

Table 1: Approaches to measuring personality-related responses in language models.

Study	Instrument and design	Main focus
Miotto et al. [4]	Itemwise HEXACO and values questionnaires	Temperature and response history
Jiang et al. [5]	MPI, personality prompting, and vignettes	Measuring and inducing trait expression
Shu et al. [6]	Binary questions with format, option, and negation changes	Consistency under prompt changes
Bodroža et al. [7]	Human-role questionnaires on two occasions	Temporal agreement
Serapio-García et al. [8]	IPIP-NEO-300, BFI-44, and varied prompts	Reliability and several forms of validity
Wang et al. [9]	Interviews and LLM-based assessment	Fidelity to assigned characters
Lee et al. [10]	Big Five and Dark Triad scenario questions	Choices in hypothetical situations
Ma et al. [11]	Word-association inventory, preprint	Sentiment associations
Present study	Three, five, or ten repeats of the full IPIP-50	Self-ratings, repeatability, and answer coverage

3 METHODS

3.1 MODELS AND QUESTIONNAIRE

The comparison covers 55 configurations from 17 developers, selected to span several model families. Artificial Analysis rankings guided selection [12], not interpretation of personality scores. Appendix A lists all configurations and trait coverage; Supplementary Table S3 gives requested identities and settings. Table S14 adds service, routing, and collection metadata. These identities refer to API labels, not independently inspected model weights. The panel does not cover every available model or leaderboard effort variant.

Initial screening covered 58 candidates. GLM 5.3 Flash, Gemini 3.1 Pro, and Step 3.7 Flash did not produce usable setup responses for this collection, leaving 55 in the main comparison. The setup records distinguish service rejections, unavailable quota, and truncated output. Setup responses are excluded from the results. Models could enter the comparison even if they marked some items not applicable; admission did not require high scores or a complete trait profile. Claude Fable 5.1 remained outside the panel.

All fifty English statements appeared together in the published sample order [1]. Every main request used the same instruction to describe the model as an AI assistant as it generally is, rather than ideally. Ratings ran from 1, Very Inaccurate, to 5, Very Accurate, with an explicit non-applicable option. Each request began a fresh study conversation without a fictional persona or previous answers. The supplement reproduces the complete prompt.

The study used three repeat-count settings (Table 2). The 39 ten-repeat configurations requested temperature 0.7 and a 1,536-token output cap. The eleven five-repeat configurations requested a 4,096-token cap, and the five three-repeat configurations requested 1,536 tokens. The latter two groups omitted temperature and requested low or disabled reasoning where supported. No generation seed was requested. These are submitted controls, not observations of internal computation.

The ten- and five-repeat schedules shuffled model order within repetition blocks. The three-repeat additions ran sequentially by model. Each planned attempt is one questionnaire administration; a permitted service-error retry can belong to the same administration. Repeat counts were resource-limited rather than precision-based, with settings fixed before each configuration's main observations. The unified trait-wise reporting rule was chosen retrospectively. Models do not recur across settings, so setting differences cannot be separated from model differences.

3.2 SCORING AND INCOMPLETE ANSWERS

A usable questionnaire response had to include an unambiguous rating or non-applicable mark for every item. Deterministic format recovery accepted explicit values in malformed wrappers or numeric strings, without inferring missing answers. It also restored one missing final brace when a normally finished bare map already contained all fifty unique item IDs and complete valid values. An independent reader checked those original values directly. Missing items, conflicting answers, out-of-range ratings, and token-limited outputs remained unusable.

The published key assigns items to five ten-item domains [2; 3]. Positive-keyed items keep their rating; negative-keyed items are scored as six minus the rating.

A domain was scored only when all ten items had numeric answers. Its score is the keyed sum divided by ten, giving a 1-5 metric. This treats category steps as equally spaced for descriptive purposes. The scores are not human-normed or calibrated as psychological distances within or between traits. Reported means describe complete-domain occasions, which may exclude some planned attempts.

Each model-domain estimate required at least three complete assessments and at least half its planned attempts. The thresholds were therefore five of ten, three of five, and three of three. The rule is a reporting choice, not a reliability guarantee. A complete profile has reportable estimates in all five domains, but need not have complete answers on every attempt.

3.3 Comparing scores and repeated answers

For each reported trait, the analysis gives the mean, sample standard deviation (SD), observed range, and complete-assessment count. The configuration is the comparison unit; item ratings and repeats are nested observations. Panel summaries give equal weight to model means and show their median, 25th and 75th percentiles (Q1 and Q3), and range. Different traits can include different models. Supplementary Table S1 lists every reported estimate.

Numeric agreement averages the share of identical numeric-answer pairs across items rated numerically at least twice. Each item receives equal weight. Applicability agreement applies the same calculation to numeric-versus-null status across all fifty items. Both measures exclude failed requests and do not adjust for chance agreement.

Exploratory checks hold item identity constant when comparing agreement, restrict profile comparisons to entirely numeric questionnaire responses, and vary the required share of complete attempts from 40% to 100%, always retaining the minimum of three. No population significance test or overall personality-quality ranking is used.

3.4 How sensitive are the comparisons?

The first check compares every possible three-assessment mean from one model with every such mean from another in the same repeat-count setting. A full-sample ordering meets the 95% stability rule when it holds in at least 95% of these comparisons. This cutoff is a descriptive reporting choice; the supplement also reports 80%, 90%, and 100% rules. Pairs with tied full-sample means or only one group for either model are excluded from stability-rate summaries. Ties between three-response averages remain in the denominator but do not preserve a strict ordering. These checks reuse recorded answers; they are not new experiments or confidence levels.

The second check changes the spacing between keyed score categories while preserving their order. Using entirely numeric questionnaires, it tests whether Agreeableness, Conscientiousness, and Intellect/Imagination remain above Extraversion under every such recoding. The supplement gives the mathematical criterion and derivation.

The third check calculates bounds for a hypothetical average over all planned item positions, unlike the reported means over complete domains. Each unknown position can range from 1 to 5, including non-applicable marks and all positions in unusable responses. If one trait's lower bound exceeds another's upper bound, their ordering holds under that assumption. No missing rating is added to the data or treated as a hidden true answer. A further check compares positive-keyed and reverse-scored negative-keyed items within fully numeric responses. These additional analyses were chosen after inspecting the data and are exploratory.

4 Results

4.1 Which traits could be scored?

The 460 main attempts produced 422 usable questionnaire responses (91.7%), including 57 recovered through format-only normalization. The other 38 outcomes comprised 23 rate-limit rejections, one service-unavailable error, nine token-limited outputs, four incomplete or invalid maps, and one interrupted request without a saved response. The four invalid maps came from Llama 3.2 1B. Three omitted item 2; the fourth included ratings outside the allowed 1-5 range. These outcomes remain in the attempt count. A non-applicable mark is a recorded answer in a usable response, but leaves its domain unscored for that attempt.

The usable responses contained 17,940 numeric ratings and 3,160 non-applicable marks, or 15.0% of 21,100 returned item positions. These item counts weight responses rather than models. They yielded 1,481 complete domain assessments. Of 275 possible model-domain combinations, 178 qualified for reporting, using 1,377 assessments. Forty-seven configurations had at least one estimate, while 22 qualified in all five domains.

Table 2: Main comparison by repeat-count setting. A usable response answers all fifty items with numbers or non-applicable marks. Setup responses are excluded.

Assessment setting	Models	Attempts per model	Total attempts	Usable responses
10-repeat	39	10	390	353
5-repeat	11	5	55	54
3-repeat	5	3	15	15
Overall	**55**	**3, 5 or 10**	**460**	**422**

Appendix A shows trait coverage for every configuration. GPT-6 Astra, GPT-6.1 Sol, Grok 4.7, and Kimi K3 qualified only for Intellect/Imagination. Claude Sonnet 5.5, Claude Opus 5.5, Gemini 3.8 Flash, and Mistral Large 4 qualified for Agreeableness, Emotional Stability, and Intellect/Imagination. MiMo-V2.6-Pro and GPT-6 Luna had usable responses but no eligible trait estimate.

Non-applicable marks were most frequent for leaving belongings around (181/422, 42.9%) and forgetting to put things back (175/422, 41.5%). Being the life of the party received 141 such marks (33.4%), and talking to different people at parties received 138 (32.7%). All five Claude Sonnet 5.5 responses marked the two belongings items and the item about talking to people at parties as not applicable, preventing Conscientiousness and Extraversion scores.

4.2 Patterns across the profiles

In 20 of the 22 complete profiles, Agreeableness, Conscientiousness, and Intellect/Imagination averages were each higher than Extraversion. Gemma 3 4B and Llama 3.2 1B were the exceptions. Their Conscientiousness averages were 2.82 and 3.15, below their Extraversion averages of 3.49 and 3.26. The same 20-model pattern held on entirely numeric questionnaires, so it did not require comparing traits answered on different occasions.

Changing the spacing of the keyed score categories left this three-trait ordering unchanged in 18 of the 22 profiles. Gemma 3 4B, Llama 3.2 1B, Command A Plus, and Granite 4 Micro did not meet this stronger rule. Their usual averages do not establish all three comparisons independently of category spacing (Supplementary Table S9).

Table 3 shows an Extraversion median of 2.83, while the other domain medians exceeded four. Different domains include different model sets, unlike the matched-profile comparison above. Emotional Stability had the widest range of means, 2.83 points, from Llama 3.2 1B at 2.17 to several configurations at 5.00.

Table 3: Distribution of eligible model means. Q1 and Q3 are the 25th and 75th percentiles. Median SD summarizes variation across complete assessments within each model.

Domain	Models	Median	Q1-Q3	Range of means	Median SD
Extraversion	29	2.83	2.44-3.17	1.56-3.83	0.29
Agreeableness	40	4.32	3.91-4.50	3.18-5.00	0.24
Conscientiousness	25	4.44	3.73-4.78	2.82-4.99	0.22
Emotional Stability	38	4.03	3.58-4.68	2.17-5.00	0.28
Intellect/Imagination	46	4.41	4.07-4.70	3.17-5.00	0.21

The groups in Figure 1 had Emotional Stability medians of 3.93 for 28 eligible ten-repeat configurations, 4.55 for seven five-repeat configurations, and 4.40 for three three-repeat configurations. No three-repeat configuration qualified for Conscientiousness. Different model compositions prevent attributing these differences to repeat count or model age.

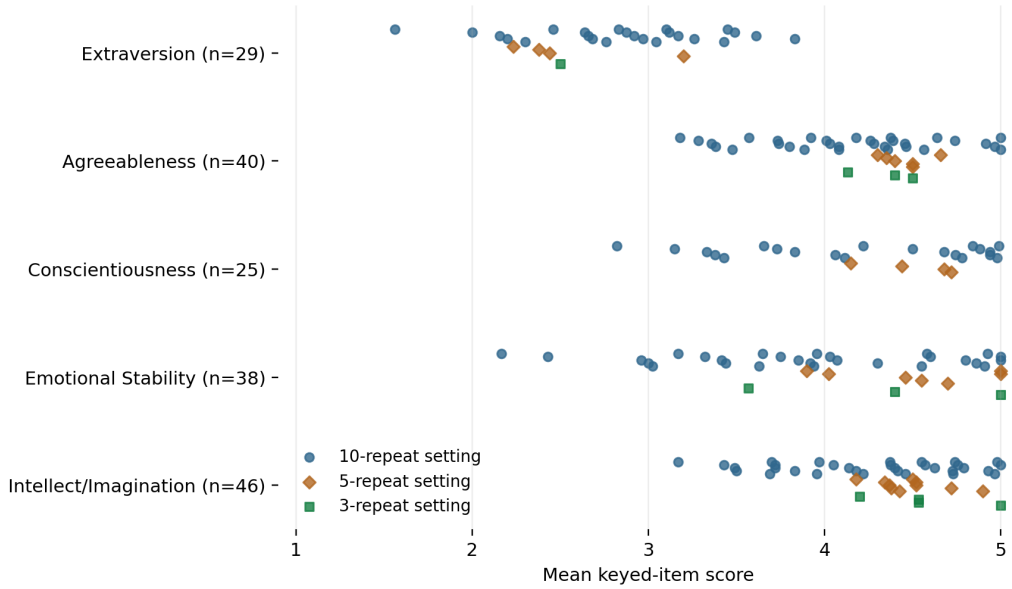


Figure 1: Distribution of all 178 eligible trait means on the 1-5 coded scale. Each point represents one configuration-domain estimate with equal model weight. Shapes and colors identify the ten-, five-, and three-repeat settings; small vertical offsets separate points. Counts beside each domain are eligible configurations. Supplementary Table S1 gives every mean, sample SD, observed range, and assessment count.

Among the 22 complete-profile models, reversed Agreeableness items averaged higher than positive-keyed items in twelve models when using fully numeric responses. The median difference within a model was 0.18 points. For Emotional Stability and Intellect/Imagination, that median difference was -0.03. Item content and keying direction were not varied separately, so this does not test a wording effect (Supplementary Table S12).

4.3 Differences between individual models

Gemma 3 4B and Gemma 3 12B each had ten complete assessments under the same requested temperature and output cap. Their Extraversion means were 3.49 and 1.56, respectively, a difference of 1.93 points. Conscientiousness ran in the opposite direction, at 2.82 and 4.68. Gemma 3 27B supplied five complete assessments for four domains but none for Conscientiousness. These observations do not support a simple progression with parameter count.

The three complete profiles in the five-repeat setting also differed. Qwen3.8 Max Prime had Extraversion 3.20 and Emotional Stability 4.46; Gemini 3.5 Flash-Lite had 2.44 and 5.00; MiniMax M3 had 2.38 and 3.90. Every mean uses five assessments.

Claude Sonnet 5.5 and Claude Opus 5.5 both averaged 4.50 in Agreeableness. Their Emotional Stability means were 4.70 and 4.40, and their Intellect/Imagination means were 4.50 and 4.20. Each reported domain had zero observed SD, over five Sonnet and three Opus assessments. Llama 4 Maverick likewise repeated its four reportable scores across three attempts, including 5.00 for Emotional Stability and Intellect/Imagination, yet never supplied a complete Conscientiousness assessment.

4.4 How consistent were the answers?

Numeric agreement ranged from 25.5% to 100% across the 53 configurations with repeatedly numeric items. This mixes different item sets and repeat counts. Among the 22 complete-profile configurations, all fifty items had repeated numeric answers, yet agreement ranged from 25.5%

for Llama 3.2 1B to 96.7% for Gemma 3 4B. These estimates used six and ten usable responses, respectively.

Fourteen items had at least two numeric answers in each of the 53 comparable configurations. Restricting agreement to that shared set reversed 186 of the 1,378 possible model-pair orderings; ties were not counted as reversals. For example, Llama 4 Scout and MiniMax M3 had 58.8% and 60.8% agreement on their own full fifty-item sets, but 66.3% and 54.3% on the fourteen shared items. The intersection contains six Intellect/Imagination, three Extraversion, three Conscientiousness, two Agreeableness, and no Emotional Stability items. It holds item identity constant at the cost of narrower content coverage (Supplementary Table S2).

GPT-6 Luna marked every item non-applicable in four of five responses and supplied 36 numeric ratings in the remaining response. MiMo-V2.6-Pro marked every item non-applicable once in three responses. Their applicability agreement was 71.2% and 34.7%, respectively. Luna had no item rated numerically twice and therefore no numeric agreement estimate.

Fifty-four of the 295 assessments contributing to reported Emotional Stability means reached 5.00 (18.3%). The corresponding shares were 50/208 for Conscientiousness (24.0%) and 0/227 for Extraversion. An Emotional Stability ceiling requires rating its two positive items as five and its eight negative items as one. It represents the keyed response pattern, not repeated selection of the same raw option.

4.5 How the reporting threshold affects comparison

Requiring complete domains on every planned attempt reduced eligible estimates from 178 to 112. Extraversion's median stayed at 2.83 and Intellect/Imagination's changed from 4.41 to 4.48. Agreeableness's median rose from 4.32 to 4.34, while its range narrowed from 1.82 to 1.43 points. Conscientiousness's median fell from 4.44 to 4.33. The changed summaries reflect different model sets, not new answers.

Strict parsing without format recovery retained 157 rather than 178 estimates. Among estimates eligible under both readings, the largest mean change was 0.05 points, for MiniMax M2.5 Intellect/Imagination. Every accepted Gemma 3 4B and Llama 3.2 1B map required recovery, so strict parsing removed both exceptions to the common profile ordering. Recovery preserves supplied values and item assignments.

4.6 Which differences survived response selection?

Gemini 3.5 Flash-Lite's Extraversion average exceeded MiniMax M3's by only 0.06 points. Comparing every possible group of three recorded assessments from each model gave 100 comparisons. These reuse five assessments per model, rather than adding 100 new observations. Gemini stayed higher in sixty, while MiniMax was higher in forty.

Qwen3.8 Max Prime's 0.82-point Extraversion difference from MiniMax M3 retained its direction in all 100 comparisons (Table 4). The Gemma 3 4B and 12B Extraversion and Conscientiousness differences also retained their directions in every comparison, although the higher-scoring model differed by trait.

Table 4: Examples of stable and unstable score differences. Difference is the first model's mean minus the second's. Each comparison uses three recorded assessments per model. Reversals put the model with the higher full-sample mean below the other. These examples had no ties. Full pair results accompany Supplementary Table S10.

Models compared (first / second)	Trait	Mean difference	Comparisons	Order reversed
Gemma 3 4B / Gemma 3 12B	Extraversion	1.93	14,400	0
Gemma 3 4B / Gemma 3 12B	Conscientiousness	-1.86	14,400	0
Qwen3.8 Max Prime / MiniMax M3	Extraversion	0.82	100	0
Gemini 3.5 Flash-Lite / MiniMax M3	Extraversion	0.06	100	40

Across the ten-repeat group, 284 of 463 evaluable Agreeableness pairs (61.3%) and 279 of 377 Emotional Stability pairs (74.0%) met the 95% stability rule, the lowest and highest domain shares. The analysis retains all 1,930 within-setting model-domain pairs, including ties and non-evaluable cases. Models with only three complete scores have one possible group and cannot support this check, regardless of observed SD. The three-repeat setting has no evaluable pairs.

4.7 What can be learned from incomplete answers?

Allowing each missing rating any value from 1 to 5 left Agreeableness, Conscientiousness, and Intellect/Imagination above Extraversion for 26 of 55 configurations. Nine of these models lacked a complete eligible profile. For the other models, observed answers either contradicted the ordering or left it unresolved under this assumption.

Kimi K3 qualified only for Intellect/Imagination. Under the hypothetical 1-5 completion assumption, its Extraversion bounds were 2.77-3.17. The lower bounds for Agreeableness, Conscientiousness, and Intellect/Imagination were 4.33, 4.03, and 4.53, all exceeding Extraversion's upper bound.

Claude Opus 5.5 had overlapping bounds for Extraversion (2.80-3.60) and Conscientiousness (3.30-4.50), leaving their ordering unresolved. Its eligible traits still had zero observed SD.

Across all 55 configurations, the median bound width was 1.12 points for Conscientiousness, 0.80 for Extraversion, and 0.00 for Intellect/Imagination. Intellect/Imagination also had the most reported estimates, covering 46 configurations. Supplementary Table S11 gives every model's limits.

5 Discussion

The twenty-of-22 pattern concerns complete profiles, covering 40% of the tested panel. Applicability and response success select this subset. Cross-domain orderings compare item responses, not calibrated psychological levels. Bodroža et al. reported related patterns [7], but different instruments, prompts, and human norms prevent direct score comparisons.

Zero observed SD does not establish deterministic output, and subset frequencies are not future-response probabilities. Completion bounds concern all planned positions, while category-spacing checks recode observed answers. Passing these checks separately does not guarantee that conclusions survive both changes together.

Human-oriented wording may affect numeric coverage. Parties and possessions may prompt literal or conversational-analogy readings, but this comparison held instructions fixed. Prompt-sensitivity and validation research [6; 8] supports separating repeatability from validity.

The sample was selected for breadth, not population representativeness. Small repeat counts, one English item order, retrospective checks, and configuration differences limit generalization. Backends and model-specific controls can differ even within a repeat-count setting. Training exposure and hidden instructions were not measured. Testing generalization requires fresh responses, controlled instructions, and behavioral tasks [9; 10].

6 Conclusion

The questionnaire revealed a common pattern in twenty of the 22 complete profiles available from 55 tested configurations. Agreeableness, Conscientiousness, and Intellect/Imagination self-ratings were higher than Extraversion in those profiles. The finding describes a selected subset under one instruction, rather than a universal model characteristic.

Some between-model differences persisted across recorded responses; others reversed. Missing answers left other comparisons unresolved. Whether questionnaire profiles predict behavior requires separate testing.

APPENDIX A. FULL MODEL COVERAGE

	E	A	C	ES	I
Qwen2.5 72B	10/10	10/10	10/10	10/10	10/10
Qwen2.5 7B	0/10	0/10	0/10	0/10	0/10
Qwen3 235B A22B	8/10	7/10	8/10	9/10	9/10
Qwen3 30B A3B	2/10	2/10	3/10	4/10	4/10
Qwen3 32B	1/10	6/10	0/10	4/10	6/10
Qwen3.8 27B	4/10	10/10	0/10	10/10	10/10
Qwen3.8 Max Prime	5/5	5/5	5/5	5/5	5/5
Nova Lite	5/10	6/10	5/10	8/10	7/10
Nova Micro	0/10	0/10	1/10	2/10	9/10
Claude Haiku 4.5	6/10	10/10	1/10	10/10	10/10
Claude Opus 5.5	0/3	3/3	0/3	3/3	3/3
Claude Sonnet 5.5	0/5	5/5	0/5	5/5	5/5
Command A Plus	10/10	10/10	10/10	10/10	10/10
Command R	5/10	4/10	0/10	1/10	1/10
Command R7B	10/10	9/10	9/10	9/10	9/10
DeepSeek V3.2	2/10	9/10	9/10	9/10	9/10
DeepSeek V4.1 Flash	3/5	2/5	1/5	4/5	5/5
Gemini 2.5 Flash-Lite	10/10	10/10	2/10	10/10	10/10
Gemini 3.5 Flash-Lite	5/5	5/5	5/5	5/5	5/5
Gemini 3.8 Flash	0/5	4/5	0/5	4/5	4/5
Gemma 3 12B	10/10	10/10	10/10	10/10	10/10
Gemma 3 27B	5/10	5/10	0/10	5/10	5/10
Gemma 3 4B	10/10	10/10	10/10	10/10	10/10
Granite 4 Micro	10/10	10/10	10/10	10/10	10/10
Llama 3.1 8B	0/10	1/10	0/10	1/10	0/10
Llama 3.2 1B	5/10	6/10	6/10	6/10	5/10
Llama 3.2 3B	10/10	6/10	10/10	10/10	10/10
Llama 3.3 70B	3/10	8/10	10/10	8/10	10/10

	E	A	C	ES	I
Llama 4 Maverick	3/3	3/3	0/3	3/3	3/3
Llama 4 Scout	10/10	10/10	10/10	10/10	10/10
Phi-4	0/10	0/10	0/10	1/10	1/10
MiniMax M2.5	1/10	8/10	0/10	1/10	8/10
MiniMax M3	5/5	5/5	5/5	5/5	5/5
MiniMax-01	10/10	10/10	10/10	10/10	10/10
Mistral Large 4	0/3	3/3	0/3	3/3	3/3
Mistral Nemo	2/10	6/10	3/10	7/10	9/10
Mistral Small 2501	4/10	7/10	4/10	10/10	10/10
Mistral Small 3.2	7/10	8/10	4/10	8/10	8/10
Kimi K2 0905	8/10	8/10	6/10	8/10	8/10
Kimi K3	2/3	2/3	1/3	2/3	3/3
Nemotron 3 Nano	10/10	10/10	10/10	10/10	10/10
Nemotron 3 Super	7/10	7/10	7/10	7/10	7/10
GPT-3.5 Turbo	10/10	10/10	10/10	10/10	10/10
GPT-4.1 mini	10/10	10/10	10/10	10/10	10/10
GPT-4.1 nano	10/10	10/10	10/10	10/10	10/10
GPT-4o mini	10/10	10/10	9/10	10/10	10/10
GPT-6 Astra	0/5	0/5	0/5	0/5	5/5
GPT-6 Luna	0/5	0/5	0/5	0/5	1/5
GPT-6.1 Sol	0/5	0/5	0/5	0/5	3/5
GPT-OSS 120B	0/10	2/10	0/10	1/10	2/10
GPT-OSS 20B	2/10	2/10	1/10	2/10	3/10
Grok 4.7	0/5	0/5	0/5	0/5	5/5
MiMo-V2.6-Pro	0/3	2/3	1/3	1/3	2/3
GLM 4.7 Flash	1/10	7/10	2/10	3/10	8/10
GLM 5.3 Prime	2/5	5/5	4/5	4/5	5/5

Figure 2: All 55 tested configurations and their answer coverage. Each cell gives complete domain assessments divided by planned attempts. E = Extraversion; A = Agreeableness; C = Conscientiousness; ES = Emotional Stability; I = Intellect/Imagination. Reported averages require five of ten, three of five, or three of three complete assessments. Zero is a coverage count, not a low trait score. Supplementary Table S4 provides the same roster as text.

DATA AND DISCLOSURE

The accompanying research package provides the responses and analysis code. Its supplement contains all reported scores, the full model roster, exact prompt, requested settings, and sensitivity tables. AI assistance was used for literature review, data collection, analysis, and writing.

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